

Xanthovar Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-152/B

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Deviations observed during the campaign were investigated and closed with no impact to product quality. The control strategy links critical quality attributes to critical process parameters identified during development. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Section 3.2.P.4 — Excipients

For the commercial formulation F-152/B, Avicel PH-200 (Microcrystalline cellulose, filler-binder) is used.

Excipient Grade Selection

The formulation development rationale for Avicel PH-200 rests on reduced sticking and picking observed during tableting trials.

The excipient control strategy was developed in accordance with ICH Q3D.

Registered Specification

Component	Function	Grade per current specification
Microcrystalline cellulose	filler-binder	Avicel PH-101
Reference		SPEC-EX-9126 Rev 07

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. No changes to the approved analytical procedures are proposed in this submission.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.